An Approach to Position Sensorless Drive for Brushless DC Motors

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Abstract— This paper describes a brushless dc motor without position sensor. Variable speed is achieved by adjusting the average motor voltage, just like chopper control of dc motors. The position sensorless drive proposed in this paper is based on detection of the conducting interval of free wheeling diodes connected in antiparallel with power transistors. This approach makes it possible to detect the rotor position over a wide speed range, especially at a lower speed. Experimental results obtained from a prototype brushless dc motor of rating 300 W are shown to confirm the validity of the sensorless drive from 45 to 2300 rpm. The starting procedure of the motor is also discussed because it is impossible to detect the rotor position at a standstill.

1 Introduction

Brushless dc motors have been put into practical use, because of high efficiency and good controllability. A brushless dc motor requires an inverter and a position sensor to perform "commutation", because a permanent magnet synchronous motor takes the place of a dc motor with brushes and commutators[2, 3]. Three Hall sensors have been used as a position sensor for the brushless dc motor. According as the motor rating decreases, the volume ratio of the Hall sensors to the motor increases, so that the miniaturization of the motor becomes hard. Furthermore, the Hall sensors need at least 8 signal wires.

In recent years, a position sensorless brushless dc motor has been studied, which produces the commutation signal from back electromotive forces in the motor[5, 6]. Since the back emf's are in proportion to the rotor speed, it is difficult to detect the back emf's in a lower speed range for a conventional sensorless drive. For that reason, the conventional sensorless drive needs a complicated starting procedure[5].

In this paper, the authors propose a position sensorless dc motor which is based on a new position detecting method. The position information is detected on the basis of the conducting state of the free wheeling diodes connected in antiparallel with power transistors, because a current flowing in a phase, in which no active drive signal is given to the positive or negative side transistor, results from the back emf's produced in the motor windings. This approach makes it possible to detect the rotor position over a wide speed range, especially at a lower speed, and to simplify the starting procedure. Experimental results obtained from a prototype sensorless drive is shown to confirm the validity of the position detecting method proposed in this paper.

2 Brushless DC Motors

Generally, a brushless dc motor consists of a permanent magnet synchronous motor which converts electrical energy to mechanical energy, an inverter corresponding to brushes and commutators, and a position sensor. Fig.1 shows the main circuit of the brushless dc motor discussed in this paper. The three-phase permanent

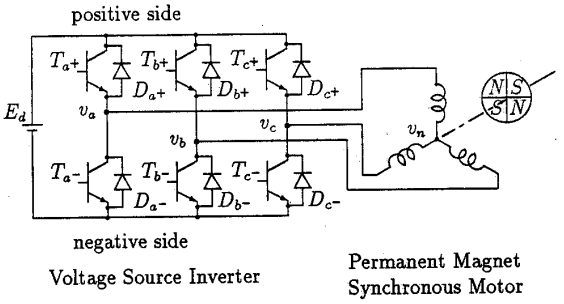


Fig.1 Brushless dc motors.

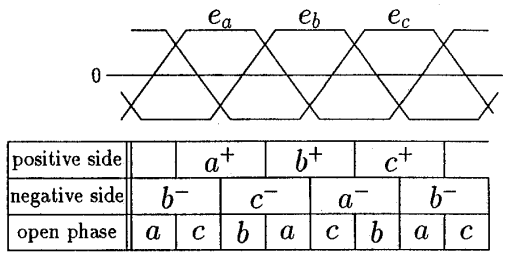


Fig.2 Relationship between drive signals and back emf's.

magnet synchronous motor has the trapezoidal back electromotive forces as shown in Fig.2. The inverter used here is the same as a conventional three-phase voltage source inverter, but the conducting interval is 120° by electrical angle as shown in Fig.2. Therefore, only two transistors, i.e. a positive side transistor in one phase and a negative side transistor in another phase, are on-state at a time. The other phase, in which no active drive signal is given to the positive or negative transistor, is called the "open phase". To continue producing the maximum torque, the commutation of inverter should be performed every 60° so that the rectangular-shaped stator current is in phase with the back emf.

3 Chopper Control

If one of the two transistors being on-state is turned on and off at a chopper frequency, the drive system enable to control the motor speed by adjusting the duty ratio, just like chopper control of dc motors. Fig.3 shows the experimental waveforms under the chopper operation. The complicated waveforms result from chopper control. Fig.3(a) and (b) are the waveforms in case that the chopper signal is given to the positive and negative transistors,

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respectively. The line current waveforms indicate the followings.

(1) The line current flows even in the period of an open phase.

(2) The beginning or ending point of the current in the open phase is the middle between the two commutations.

In the proposed brushless dc motor, the alternate chopper shown in Fig.4 is adopted to symmetrize the current waveforms and to disperse the switching losses to the positive and negative side transistors. Fig.3 (c) shows experimental waveforms in the case of the alternate chopper. The current in the open phase always begins to flow at the middle point. In the proposed sensorless drive described in next section, the rotor position is detected at this point.

4 Position Detecting Method

Fig.5 shows the simplified circuit in which the active signal is given to T_a+ and T_b-. In this case, the chopper signal is applied to T_a+, and the c-phase is open. If T_a+ is on-state, the dc link voltage increases the main current i. If T_a+ is turned off, i continues to flow through the free wheeling diode D_a- and decreases. Then, the c-phase voltage is given by

$$v_c = e_c + \frac{V_{CE} - V_F}{2} - \frac{e_a + e_b}{2} \tag{1}$$

where V_{CE} and V_F are the forward voltage drop of the transistors and diodes, respectively(Appendix). Eq.(1) holds good even in transient states, because any motor constants are not included in (1) although the resistances and inductances of the motor windings are taken into consideration in the derivation of (1). The conducting condition of the diode D_c- is given by

$$v_c < -V_F. \tag{2}$$

Substituting (1) into (2) derives the following:

$$e_c - \frac{e_a + e_b}{2} < -\frac{V_{CE} + V_F}{2}. \tag{3}$$

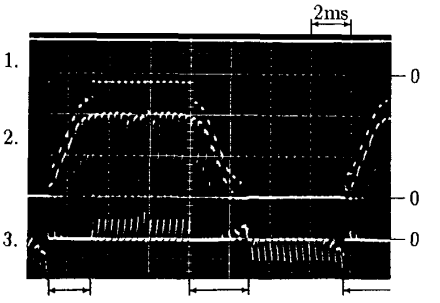
Since the back emf's are trapezoidal, e_a+e_b is approximately zero near the zero point of e_c where the left and right hands of (3) are equal. Therefore, the conducting condition of D_c- is given by

$$e_c < -\frac{V_{CE} + V_F}{2}. \tag{4}$$

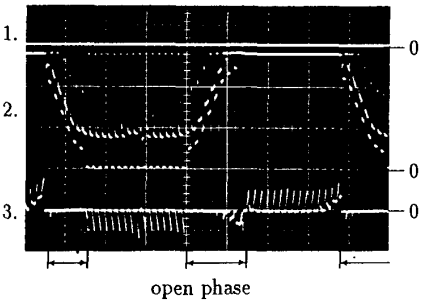
Generally, V_{CE} and V_F are enough smaller than the back emf's. When the c-phase back emf e_c becomes negative, the open phase current flows through the negative side diode D_c-.

Table 1 shows the inverter modes, and Fig.6 shows a diagram of the mode sequence. The instant of the first appearance of mode 4 is the zero point of e_c. The rotor position can be detected at this time by detecting whether D_c- is conducting or not. This point is leading from the next commutation by 30°. The commutation signal is given through a digital phase shifter using counters as shown in Fig.7. The detection is stopped right after the previous commutation, because of the main current which flowed through T_c+ before the previous commutation.

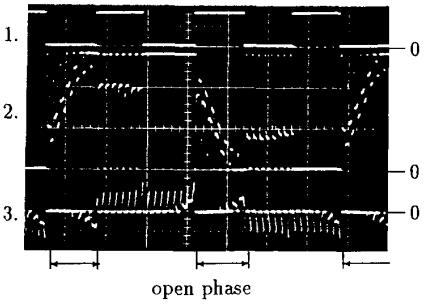
Fig.8 shows a circuit to detect whether the free wheeling diodes are conducting or not. The resistors and diodes are connected to the comparators for voltage clamping. The reference voltage V_{ref} is slightly smaller than the forward voltage drop of the free wheeling diodes V_F. The detecting circuit needs two isolated



open phase
(a) positive-side chopper



open phase
(b) negative-side chopper



open phase
(c) alternate chopper

1. chopper selecting signal 5V/div
   H:positive-side chopper, L:negative-side chopper
2. output phase voltage 50V/div
3. line current 0.2A/div

Fig.3 Experimental waveforms under chopper operation (no-load).

Table 1 Inverter modes.

mode	1	2	3	4	5	6
chopper	ON		OFF			
main current	$i > 0$				$i=0$	
open phase current	i_c=0	>	=	>	=	>
T_a+	ON		OFF			
D_a+	OFF		ON		OFF	
D_c-	OFF	ON	OFF	ON	OFF	ON

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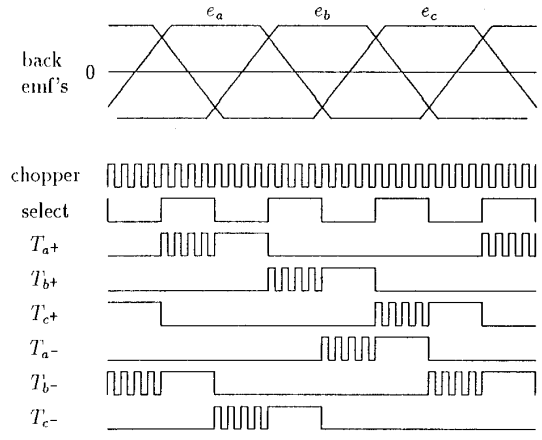


Fig.4 Drive signals under alternate copper.

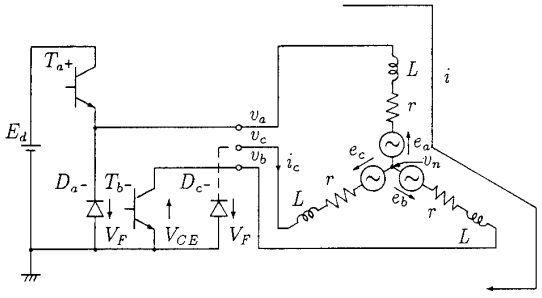


Fig.5 Simplified circuit in which active signal is given to T_{a+} and T_{b-}.

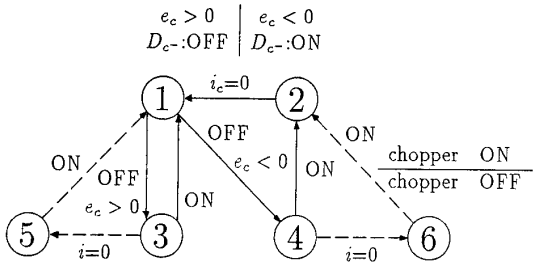


Fig.6 diagram of mode sequence.

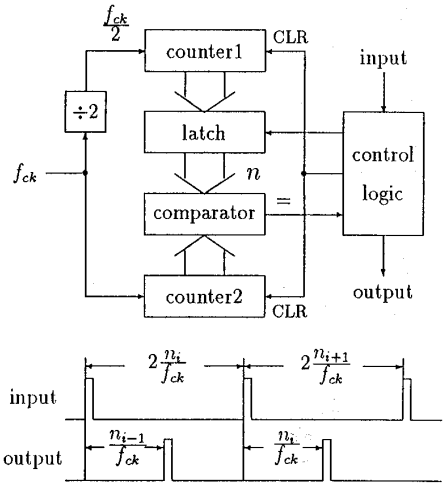


Fig.7 Digital phase shifter.

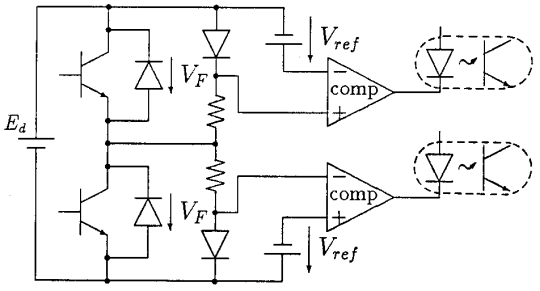
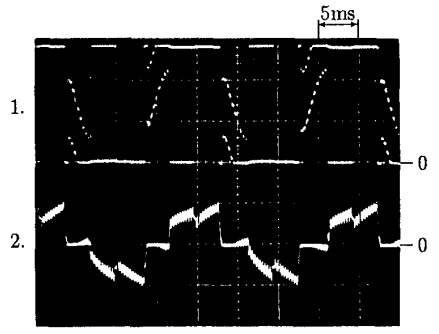
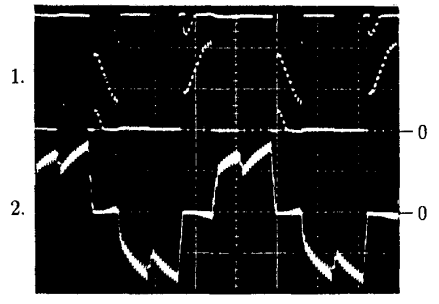


Fig.8 Detecting circuit connected to diodes.

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(a) N=1527 rpm, duty 60%



(b) N=1314 rpm, duty 60%

1. phase voltage 50V/div
2. phase current 1A/div

Fig.9 Experimental waveforms of the sensorless drive (on-load).

power supplies, but it is not influenced by the dc link voltage variation.

Fig.9 shows experimental results of the proposed position sensorless drive in steady states. Stable and good driving performance is attained up to 2300 rpm by the proposed position detecting method. The detecting characteristics are invariant with the load, because of no variation of current in the open phase as shown in Fig.9.

5 Starting Procedure

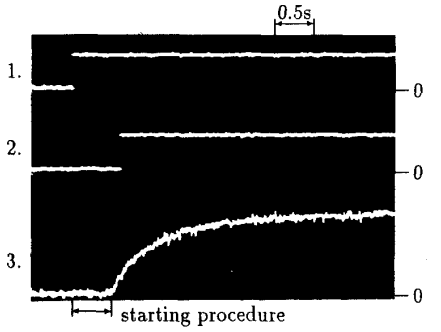
The amplitude of the back emf's is proportional to the rotor speed. Since the open phase current results from the back emf's, it is impossible to detect the rotor position at a standstill. Then a suitable starting procedure is necessary to the position sensorless brushless dc motor drive.

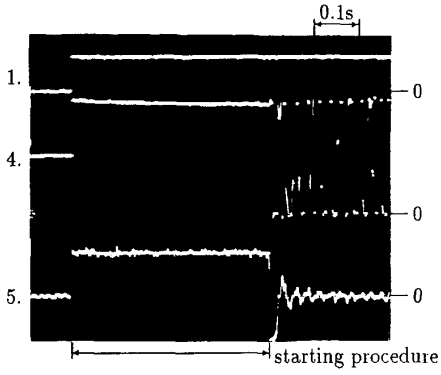
The conducting condition of the free wheeling diode is given by (4). To detect the open phase current, the open phase back emf has to be greater than $(V_{CE} + V_F)/2$ at least. The back emf coefficient of the motor used in this experimental system is 27.7 mV/rpm (phase crest value), and $V_{CE}+V_F$=1.2V. Therefore, the minimum detectable speed is theoretically given by 22 rpm. Since this value is very small, the authors propose such a simple starting procedure as the following.

First of the procedure, arbitrary two phases are exited for a preset time at the rated current of the motor. The rotor turns to the direction corresponding to the exited phases. Next, the commutation signal which advances the switching pattern by 120° is given, and then the open loop commutation is switched to the sensorless drive proposed in the previous section at 45 rpm. After the next commutation, the position sensorless drive is attained. Note that the phase shifter is not used in a speed range below 500 rpm. The reasons are the followings:

(1) The output signals from the detecting circuit connected to the free wheeling diodes have some delay in the speed range.

(2) The position sensorless drive may be unstable, because the phase shifter acts as a delay element of which the delay time varies in inverse proportion to the rotor speed.





1. starting signal 5V/div
2. phase shifter 5V/div
   H: use, L: no use
3. rotor speed 1000rpm/div
4. phase voltage 50V/div
5. phase current 2A/div

Fig.10 Starting characteristics (N=0 → 2000 rpm).

Fig.10 shows a starting characteristic. The starting signal begins the starting procedure. After the preset time of 0.5 s, the excellent starting performance is attained by the proposed starting procedure.

6 Conclusion

This paper has presented a position sensorless brushless dc motor, in which the position information is given on the basis of the conducting state of free wheeling diodes in an open phase.

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The open phase current under chopper operation results from the back emf's produced in the motor windings. Therefore, this is considered as an indirect detection of the back emf's through the free wheeling diodes. This approach is characterized by its sophisticated position detection, and makes it possible to detect the rotor position over a wide speed range from 45 to 2300 rpm, compared with the conventional method which directly detects the back emf's. The capability of position sensorless drive at the low speed of 45 rpm makes a great contribution to simplify the starting procedure. Experimental results obtained from a prototype motor of rating 300 W have shown the validity of the proposed system.

References

[1] Jaime De La Ree and Nady Boules. "Torque Production in Permanent-Magnet Synchronous Motors". In *IEEE/IAS Annual Meeting Conf. Rec.*, pages 15–20, 1987.

[2] Yoshihiro Murai, Yoshihiro Kawase, Kazuharu Ohashi, Kazuo Nagatake, and Kyugo Okuyama. "Torque Ripple Improvement for Brushless DC Miniature Motors". In *IEEE/IAS Annual Meeting Conf. Rec.*, pages 21–26, 1987.

[3] Pragasen Pillay and Ramu Krishnan. "Modeling, Simulation, and Analysis of Permanent-Magnet Motor Drives, Part I: The Brushless DC Motor Drive". *IEEE Trans. on Ind. Appli.*, 25(2):274–279, March/April 1989.

[4] Jacques Davoine, Robert Perret, and Hoang Le-Huy. "Operation of Self-Controlled Synchronous Motor without a Shaft Position Sensor". *IEEE Trans. on Ind. Appli.*, IA-19(2):217–222, March/April 1983.

[5] K. Iizuka, H. Uzuhashi, M. Kano, T. Endo, and K. Mohri. "Microcomputer Control for Sensorless Blushless Motor". *IEEE Trans. on Ind. Appli.*, IA-21(4):595–601, May/June 1985.

[6] Ray-Lee Lin, Ming-Tzwu Hu, and Ching-Yuan Lee. "Using Phase-Current Sensing Circuit as the Position Sensor for Brushless DC Motor without Shaft Position Sensor". In *IEEE/IES IECON'89 Conf. Rec.*, pages 215–218, 1989.

APPENDIX

Derivation of (1)

The equation of the loop through D_a- and T_b-, which is shown in Fig.5, is given by

$$V_F + L\frac{di}{dt} + ri + e_a - e_b + ri + L\frac{di}{dt} + V_{CE} = 0. \qquad (5)$$

Therefore, the voltage drop of the motor resistance and inductance is the following:

$$ri + L\frac{di}{dt} = -\frac{e_a - e_b}{2} - \frac{V_{CE} + V_F}{2}. \qquad (6)$$

The potential of the motor neutral point v_n, which is shown in Fig.5, is given by

$$\begin{aligned} v_n &= V_{CE} + ri + L\frac{di}{dt} - e_b \\ &= -V_F - ri - L\frac{di}{dt} - e_a. \end{aligned} \qquad (7)$$

Substitution (6) into (7) gives the following equation:

$$v_n = \frac{V_{CE} - V_F}{2} - \frac{e_a + e_b}{2}. \qquad (8)$$

Since the c-phase terminal voltage v_c equals $e_c + v_n$, the equation (1) is derived as the following:

$$v_c = e_c + v_n = e_c + \frac{V_{CE} - V_F}{2} - \frac{e_a + e_b}{2}. \qquad (1)$$

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